

1. Introduction to AR/VR Characteristics

AR/VR systems aim to create **realistic, smooth, and immersive experiences**.

- Performance depends on **visual, computational, and sensory parameters**.
- Poor system characteristics can cause **motion sickness and discomfort**.



Transition Question:

Why do some VR experiences feel realistic while others cause dizziness?

2. Key AR/VR Characteristics

Introduction to Field of View

What is Field of View?

- Field of View (FOV) refers to the **angular area of the environment visible to the user at one time**.
- Measured in **degrees (°)**.

Simple Analogy:

- Looking through a **keyhole** → small FOV



- Looking through a **window** → large FOV



 Key Idea:

The wider the FOV, the more “real” the environment feels.

2. Human Field of View vs AR/VR FOV

Human Visual Field:

- Horizontal FOV $\approx 180^\circ$
- Vertical FOV $\approx 135^\circ$
- Peripheral vision plays a major role in motion awareness.

AR/VR Device FOV:

- Typical VR headsets: 90° – 120°



- AR smart glasses: 40°–60°



 **Discussion:**
Why do current devices not match human FOV?

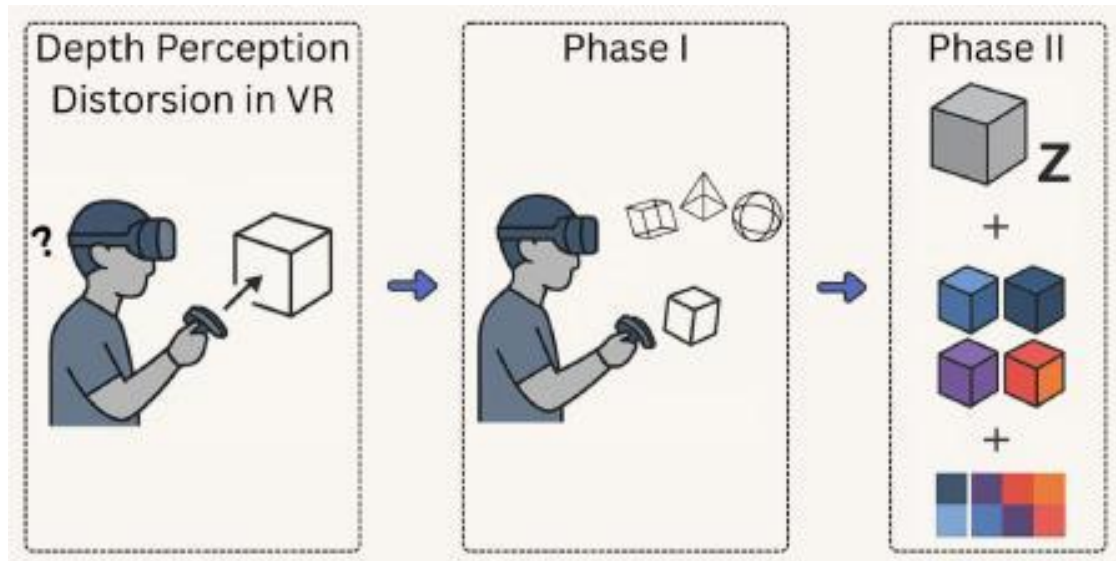
3. Role of FOV in Immersion

Relationship Between FOV and Immersion:

- FOV directly affects the **sense of presence**.
- Wider FOV reduces awareness of the real world.

High FOV Effects:

- Increased realism
- Better depth perception
- Stronger spatial presence



Example:

Driving simulation feels more realistic with wide side vision.

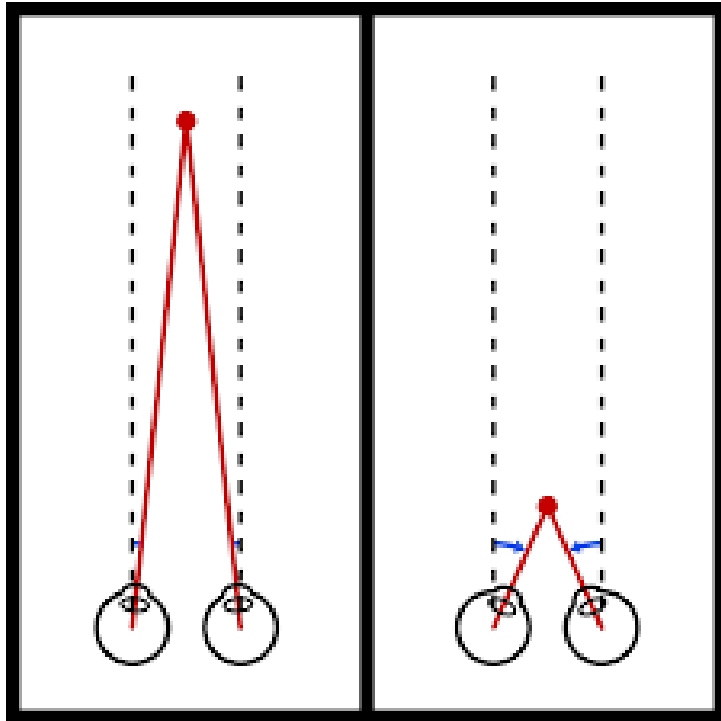
4. Impact of Narrow FOV

Effects of Narrow FOV:

- Tunnel vision effect
- Reduced situational awareness
- Lower immersion
- Increased discomfort

Common Problems:

- Breaks presence
- User feels “boxed in”
- Can cause motion sickness



📌 **Real Example:**
Early VR headsets had limited FOV causing discomfort.

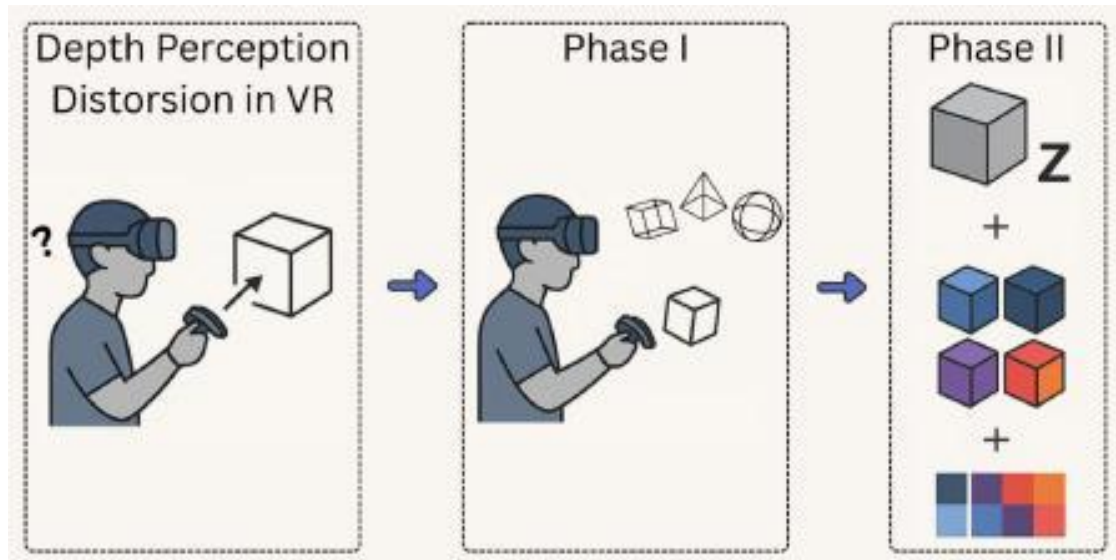
5. Impact of Wide FOV

Benefits of Wide FOV:

- Natural viewing experience
- Better motion perception
- Enhanced realism

Challenges:

- Requires higher rendering power
- Increases GPU load
- Optical distortion issues



Introduction to Latency

What is Latency?

- Latency is the **time delay** between a user's physical movement and the system's **visual response**.
- Measured in **milliseconds (ms)**.

Simple Example:

- Turn your head → image updates **after a delay** → discomfort.

📌 Key Point:

In VR, even small delays are noticeable and harmful.

2. Motion-to-Photon Latency

Definition:

Motion-to-photon latency is the **total time from user motion (head/hand movement) to the updated image appearing on the display**.

Acceptable Limits:

- Ideal: < 20 ms
- Noticeable discomfort: > 30 ms
- Severe motion sickness: > 50 ms

📌 Why it matters:

Human brain expects immediate visual feedback to motion.

3. Importance of Low Latency in AR/VR

Effects of Low Latency:

- Smooth motion
- Accurate tracking
- Strong sense of presence
- Reduced motion sickness

Effects of High Latency:

- Motion sickness (VR nausea)
- Eye strain
- Loss of immersion
- Disorientation



📌 Real-world example:

VR roller-coaster with delay causes dizziness.

4. Sources of Latency in AR/VR Systems

A. Sensor Delay (5 minutes)

Definition:

Time taken by sensors to detect motion.

Sources:

- Gyroscope
- Accelerometer
- Camera (in AR)

Impact:

Slow sensor data → delayed tracking.

B. Processing Delay (5 minutes)

Definition:

Time taken by the system to process sensor data and render graphics.

Includes:

- Tracking algorithms
- Physics calculations
- Rendering pipeline

Impact:

Heavy computation increases latency.

C. Display Refresh Delay (5 minutes)

Definition:

Time taken by the display to update pixels.

Factors:

- Refresh rate
- Pixel response time

Example:

Low refresh rate displays increase motion lag.

1. Introduction to Frame Rate

What is Frame Rate?

- Frame rate is the **number of images (frames) displayed per second**, measured in **Frames Per Second (FPS)**.
- Higher FPS → smoother visual experience.

Simple Analogy:

- Flipbook animation
 - Few pages → jerky motion
 - Many pages → smooth motion



Key Idea:

VR requires much higher frame rates than traditional video or gaming.

2. Frame Rate Requirements in AR/VR

Standard Frame Rates:

- Movies: ~24 FPS
- Traditional games: 30–60 FPS
- **VR systems: minimum 90 FPS**

Why VR Needs High FPS:

- Head movements are fast and continuous.
- Any delay or stutter breaks immersion.
- Lower FPS causes **visual judder**.



Visual Judder:

A shaking or stuttering effect during motion.

3. Impact of Low Frame Rate

Problems Caused by Low FPS:

- Jerky motion
- Motion sickness
- Eye strain
- Loss of realism

Psychological Effect:

- Brain receives inconsistent motion cues.
- Causes discomfort and nausea.



Example:

Turning head in VR but image updates slowly.

4. Benefits of High Frame Rate

Effects of High FPS:

- Smooth motion tracking
- Natural movement perception
- Reduced eye strain
- Increased sense of presence

User Experience Improvement:

- Better immersion
- Longer comfortable usage time



Engineering Insight:

High FPS is as important as high resolution in VR.

5. Frame Rate vs Hardware Performance

Factors Affecting Frame Rate:

- GPU capability
- Rendering complexity
- Scene geometry
- Lighting and shaders

Trade-off:

- Higher graphics quality → lower FPS
- VR prioritizes **FPS over visual effects**



Design Principle:

“Stable 90 FPS is better than ultra-realistic graphics.”

3. Hardware Requirements for AR/VR

Introduction to AR/VR Hardware

AR and VR experiences depend heavily on **specialized hardware**.

- Hardware determines:
 - Level of immersion
 - Tracking accuracy
 - Responsiveness
 - User comfort
- VR typically requires **dedicated devices**, while AR can work on **consumer hardware**.



Key Idea:

Better hardware → smoother, more immersive experience.

2. Hardware Requirements for Virtual Reality (VR)

A. Head-Mounted Display (HMD) (3 minutes)

Function:

- Displays stereoscopic 3D visuals.
- Covers user's eyes completely.

Key Features:

- High resolution
- Wide Field of View
- High refresh rate

 Role:

Creates full immersion by blocking the real world.

B. Motion Sensors: Gyroscope & Accelerometer (3 minutes)

Gyroscope:

- Measures rotational movement (head turning).

Accelerometer:

- Measures linear motion.

 Role:

Tracks head movement in real time for correct viewpoint updates.

C. Controllers / Data Gloves (3 minutes)

Function:

- Allow user interaction with virtual objects.

Examples of Interaction:

- Grabbing
- Pointing
- Pressing virtual buttons

 Role:

Enables natural and intuitive interaction.

D. High-Performance GPU (2 minutes)

Function:

- Renders complex 3D graphics at high frame rates (≥ 90 FPS).

 Importance:

Low GPU performance causes lag and motion sickness.

E. High-Speed CPU (2 minutes)

Function:

- Handles physics, AI, tracking, and system logic.

 Importance:

Ensures real-time response and synchronization.

F. Audio System (3D Sound) (2 minutes)

Function:

- Provides spatial audio cues.

 Role:

Enhances realism and directional awareness.

VR Hardware Examples (2 minutes)

- Standalone VR headsets
 - PC-tethered VR systems
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3. Hardware Requirements for Augmented Reality (AR)

A. Smartphone / AR Glasses (3 minutes)

Function:

- Primary computing and display device.

**Role:**

Displays virtual objects over the real world.

B. Camera (3 minutes)**Function:**

- Captures real-world environment.

**Role:**

Used for tracking, object recognition, and overlay placement.

C. GPS (Location-Based AR) (2 minutes)**Function:**

- Provides geographic position.

**Use Case:**

Navigation and outdoor AR applications.

D. Depth Sensors (2 minutes)**Function:**

- Measures distance between objects and device.

**Role:**

Improves object placement and occlusion handling.

E. IMU Sensors (2 minutes)**Components:**

- Gyroscope
- Accelerometer
- Magnetometer

**Role:**

Tracks orientation and movement of the device.

AR Hardware Examples (1 minute)

- Mobile-based AR
 - Smart glasses-based AR
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4. VR vs AR Hardware Comparison

Aspect	VR	AR
Device Type	Dedicated headset	Smartphone / Glasses
Immersion	Full	Partial
Sensors	Advanced tracking	Camera + IMU
GPU Requirement	Very High	Moderate
Mobility	Limited	High

4. 1. Introduction to AR/VR Software Stack

- AR/VR systems are **software-intensive**.
- Software controls:
 - o Rendering quality
 - o Tracking accuracy
 - o Interaction
 - o System responsiveness
- Hardware alone cannot deliver immersion without **optimized software**.

**Key Idea:**

Smooth AR/VR experiences depend more on software efficiency than visual effects.

2. Operating System (5–8 minutes)

Role of Operating System:

- Manages hardware resources
- Handles memory, scheduling, and I/O
- Supports real-time processing

Common OS for AR/VR:

- Desktop OS (for PC-VR)
- Mobile OS (for mobile AR)
- Headset-specific OS

Importance:

Low-level OS optimizations reduce latency and improve performance.

3. Rendering Engines

What is a Rendering Engine?

A rendering engine generates **real-time 3D graphics** and manages scenes, lighting, physics, and animation.

Common Engines:

- Unity
- Unreal Engine

Responsibilities:

- Scene rendering
- Physics simulation
- Lighting and shading
- Animation handling

Why Engines Matter in AR/VR:

- Must maintain **90+ FPS**
 - Must support **stereoscopic rendering**
 - Optimized for real-time interaction
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4. Software Development Kits (SDKs)

What are SDKs?

SDKs provide **libraries, tools, and APIs** to build AR/VR applications.

Common AR/VR SDKs:

- Mobile AR SDKs
- Cross-platform XR SDKs

Functions:

- Motion tracking
- Plane detection
- Object anchoring
- Gesture recognition



Benefit:

SDKs simplify complex AR/VR development tasks.

5. Drivers and APIs

Drivers:

- Enable communication between hardware and software
- Control displays, sensors, and controllers

APIs:

- Provide standard interfaces for developers
- Ensure cross-platform compatibility



Importance:

Efficient drivers and APIs minimize system overhead and latency.

6. Tracking & Sensor Fusion Software

Tracking Software:

- Detects head, hand, and device movement
- Uses cameras and sensors

Sensor Fusion:

- Combines data from multiple sensors

- Improves accuracy and stability



Example:

Combining gyroscope + accelerometer + camera data for precise tracking.

7. Key Requirement: Real-Time Rendering & Low Latency

Why It Is Critical:

- Prevents motion sickness
- Maintains immersion
- Ensures accurate interaction

Software Responsibilities:

- Fast data processing
 - Predictive tracking
 - Efficient rendering pipelines
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5. Recap & Discussion

Key Takeaways

- Frame rate controls smoothness.
- VR requires ≥ 90 FPS.
- Low FPS causes judder and discomfort.
- High FPS improves comfort and immersion.
- FOV affects immersion
- Latency affects comfort
- Frame rate affects smoothness
- Hardware + software together define experience